

Data Journalism

Session 2: Global models of data journalism

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Data journalism in the Global South

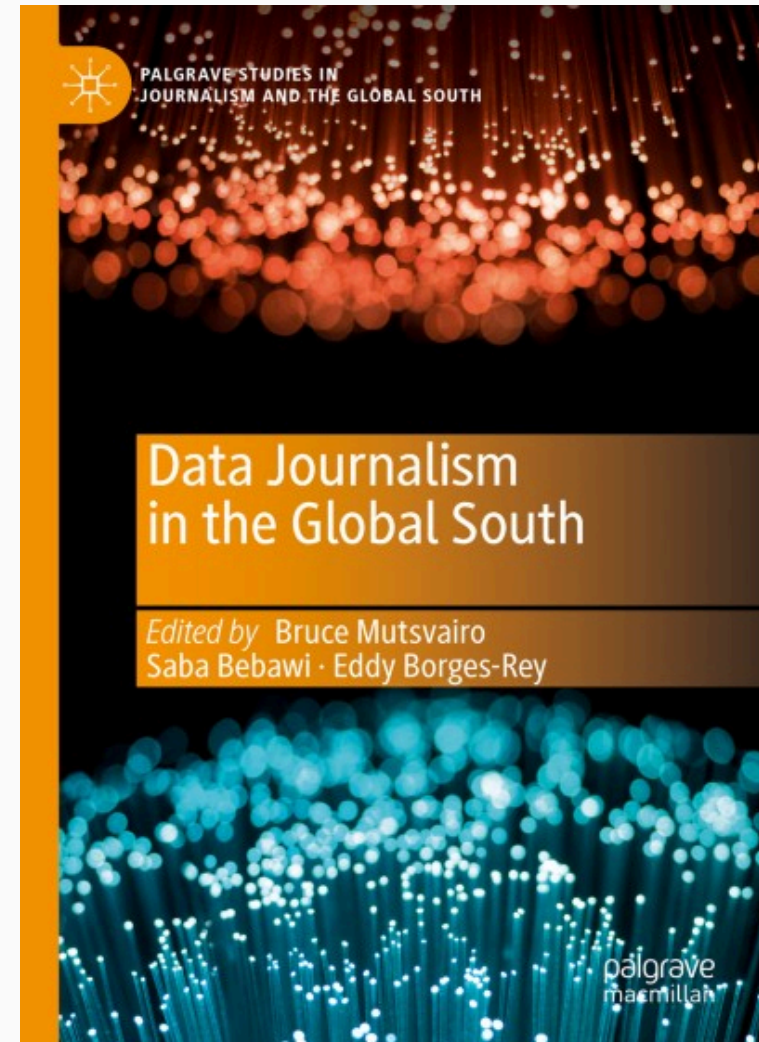
Why study data journalism beyond the West?

The problem

- Most data journalism research focuses on Western newsrooms
- Different contexts create different challenges and opportunities
- Innovation often happens at the periphery

Mutsvairo, Bebawi & Borges-Rey (2020)

- How data journalism develops in non-Western contexts
- The role of infrastructure, politics, and resources
- Alternative models and innovations

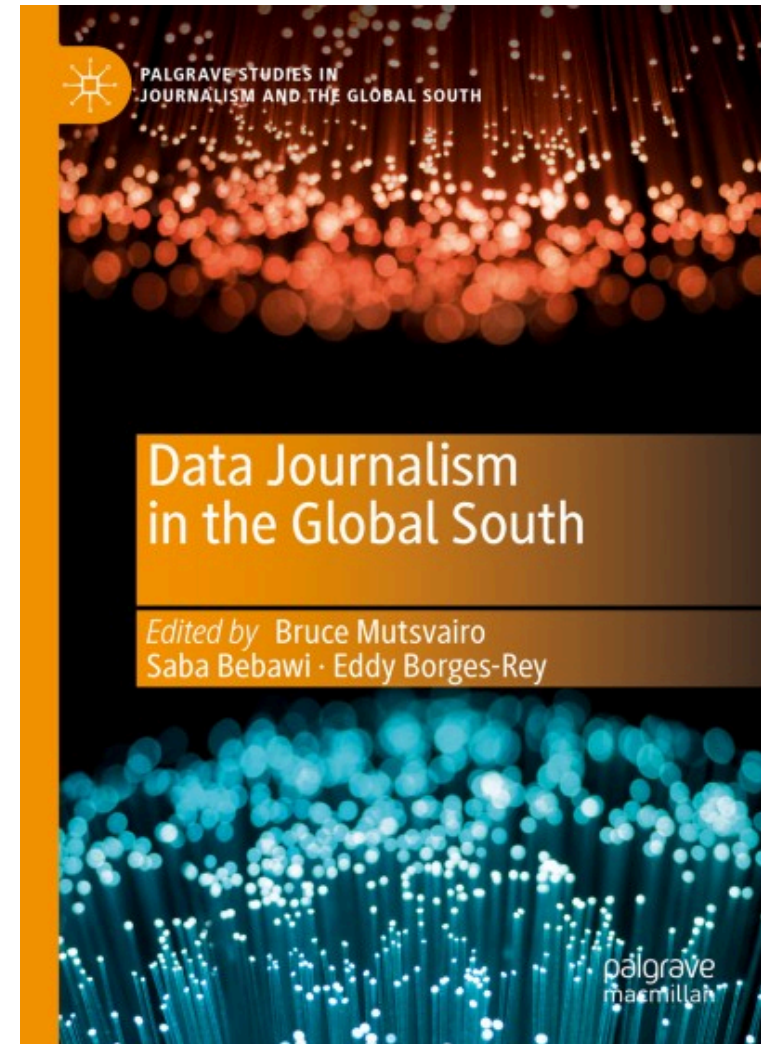


Group discussion

In groups of 3, give each other a brief summary of one chapter you read focusing on 1-2 key take-aways or interesting facts your learned.

Then, discuss the following questions in your group:

1. What are the main challenges facing data journalism in the Global South? How do they differ from Western contexts?
2. What unique opportunities or innovations have emerged in these contexts?



Key insights from the Global South

Common challenges

Infrastructure limitations:

- Limited access to open data
- Weak freedom of information laws

Resource constraints:

- Small newsrooms with limited budgets for investigative work
- Few journalists trained in data skills

Political obstacles:

- Government restrictions on information
- Threats to journalists' safety
- Censorship and self-censorship

Unique opportunities

Innovation from necessity:

- Collaboration with civic tech organizations
- Mobile-first approaches

Different priorities:

- Focus on accountability in fragile democracies
- Exposing corruption as core mission
- Building trust in divided societies

Global networks:

- Cross-border investigations (Panama Papers, Paradise Papers)
- Skill-sharing across regions
- Building transnational communities

Case study: La Nación Data (Argentina)

La Nación Data

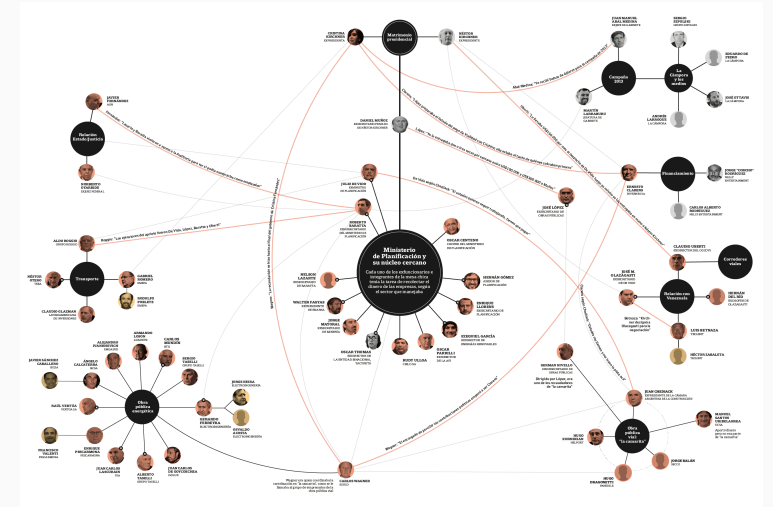
Founded: 2012 within Argentina's leading newspaper | **Model:**
Pioneered data-supported investigative journalism in Latin America

The Corruption Notebooks (2018)

What they got: Handwritten notebooks from the driver of Roberto Baratta, Secretary under the Kirchner administrations (2003-2015). The notebooks documented every bribe delivery route for a decade.

The investigation: Team spent 7 months digitizing 8 notebooks before going public; cross-checked all payments against tenders and public contracts; built network visualizations showing web of corruption

The impact: 73 people prosecuted, including fmr. President Cristina Kirchner, 40 imprisoned, 40 confessed, \$600 million seized



The investigation (Wayback machine) |
La Nación Data | Wikipedia article |
Online Journalism Award page

Case study: Caixin (China)

Caixin Media

Founded: 2009 by editor-in-chief Hu Shuli | **VisLab:** Established 2013 by Zhimin Huang ("first person to do data journalism in China") |

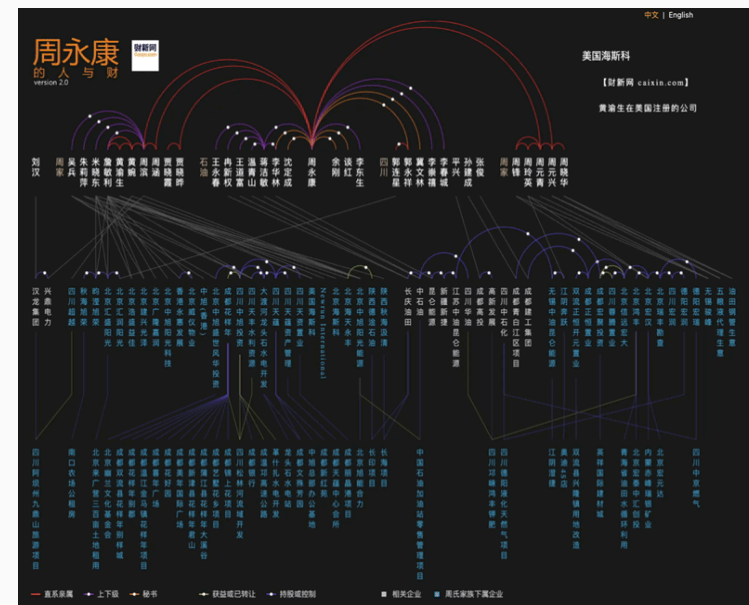
Awards: 2018 GEN Award for Best Large Data Journalism Team (first Chinese outlet)

Zhou Yongkang case (2014)

The challenge: Zhou was former head of China's security services, being investigated for corruption; Caixin wrote 60,000-word investigative story but worried "people might get tired reading"

The solution: Created network visualization showing Zhou's web of power; mapped connections between officials and companies;

Impact: 4 million hits within one week, 20 million social media views; "considered a turning point in Chinese journalism"; created appetite for visual stories that didn't exist before



Zhou's Power Base (Chinese) | Caixin Data News | GIJN article

A strategy for authoritarian contexts:
Only use publicly available data; focus on business/finance/environment not direct political criticism; show patterns not individuals

What Western journalism can learn

Beyond the technology hype

Data and open source investigative journalism in the Global South reveals that **data-informed journalism is not necessarily about sophisticated tools**, but about:

- Working with limited resources creatively
- Capitalizing on communities and networks
- Enriching more classical investigative techniques with data

Alternative models of practice

Cross-institutional collaboration: Collaborative networks execute larger investigations that single traditional media outlets cannot or will not do

Mobile-first: Innovation driven by how audiences actually access information



Discuss

1. **"So what":** What's the key take-away? And what role does the data play for establishing it?
2. **Sources & access:** What data did they use and how did they get it? Could you reproduce their analysis? What's missing?
3. **Design and storytelling:** Does their scrollytelling approach work for you? Are there flaws in visual and narrative design that you fix? How?



Source Bellingcat 2024

Open Source Intelligence (OSINT)

What is Open Source Intelligence?

Definition

"Open Source Intelligence (OSINT) is the practice of gathering, analyzing, and disseminating information from publicly available sources to address specific intelligence requirements.."

— Esteban Borges, RecordedFuture.com

For journalism: Using publicly available information—social media, satellite imagery, public records—to investigate events and hold power accountable.

Key principle: Anyone with internet access can verify the claims.

OSINT sources

- Social media posts (Twitter/X, Facebook, VK, Telegram)
- Satellite/aerial imagery (Google Earth, Planet Labs)
- Public databases and registries
- Flight tracking (Flightradar24)
- Webcams and dashcams
- Leaked documents
- Digital archives

Why OSINT matters

- Traditional sources blocked or unreliable
- Real-time documentation of events
- Verification of official narratives
- Accountability when authorities fail

The OSINT toolkit

Geolocation tools

- **Google Maps / Street View:** Matching landmarks
- **Google Earth / Satellite imagery:** Terrain, shadows, timestamps
- **SunCalc:** Determining time from shadows
- **Yandex Maps:** Often better for Eastern Europe/Russia

Social media tools

- **Social media search/APIs:** Finding relevant posts
- **InVID / WeVerify:** Video verification
- **Reverse image search:** Finding original sources

Verification techniques

- **Geolocation:** Matching photos/videos to real locations
- **Chronolocation:** Determining when content was created
- **Image forensics:** Detecting manipulation
- **Cross-referencing:** Multiple sources confirming facts
- **Metadata analysis:** Extracting hidden information

Presentation tools

- **ArcGIS StoryMaps:** Interactive timelines with map
- **TimelineJS:** Narrative timelines
- **Google My Maps:** Custom annotated maps

Case study: Bellingcat and MH17

The tragedy

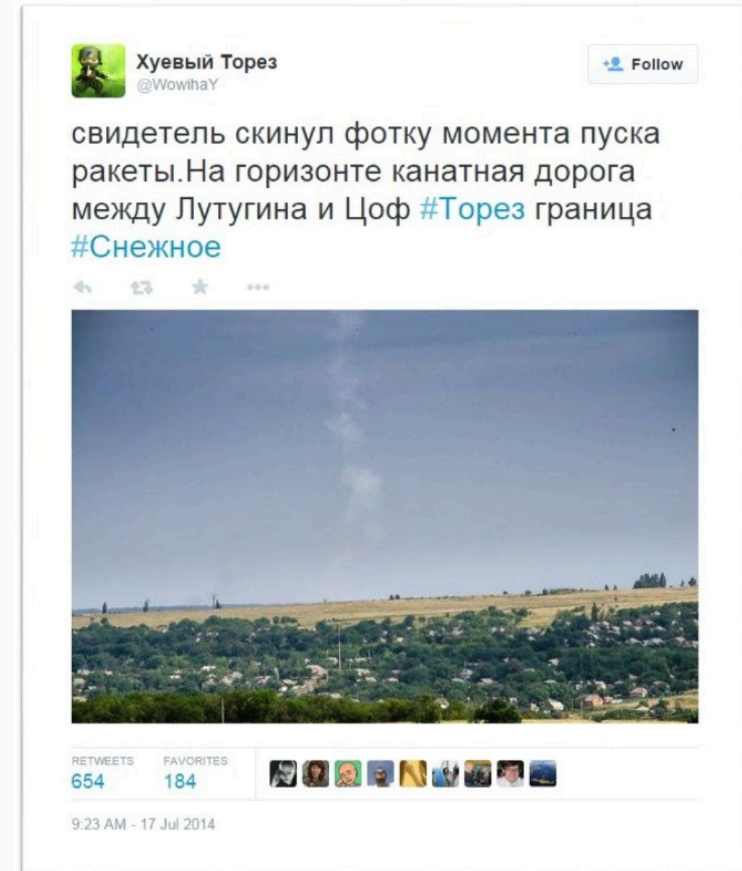
July 17, 2014: Malaysia Airlines Flight 17 shot down over Eastern Ukraine

- 298 people killed
- Boeing 777 from Amsterdam to Kuala Lumpur
- Crashed near Hrabove in conflict zone

The information war

Multiple competing narratives:

- Ukraine blamed Russian-backed separatists
- Russia blamed Ukrainian military
- Separatists claimed it was Ukrainian fighter jet
- Massive disinformation campaign



Translation: "A witness sent a photo of the moment of the missile launch. On the horizon [there are] power lines between Lutuhyne and Tsof #Torez border #Snizhne"

Bellingcat's investigation approach

What is Bellingcat?

Founded: 2014 by Eliot Higgins **Mission:** Open-source investigative journalism **Key principle:** Radical transparency—show your work

The MH17 investigation

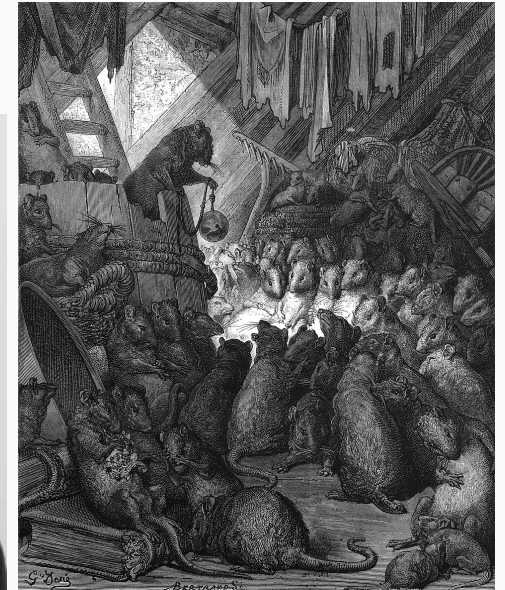
Core finding:

MH17 was shot down by a Russian Buk missile launcher transported from Russia to Ukraine and back

Evidence trail:

- Social media posts showing the Buk
- Satellite imagery
- Vehicle identification numbers
- Analysis of blast patterns

bellingcat



Left: Eliot Higgins, Bellingcat founder

Right: Gustave Doré, La Fontaine's fable (1868)

How they did it: The Buk trail

Step 1: Initial sightings

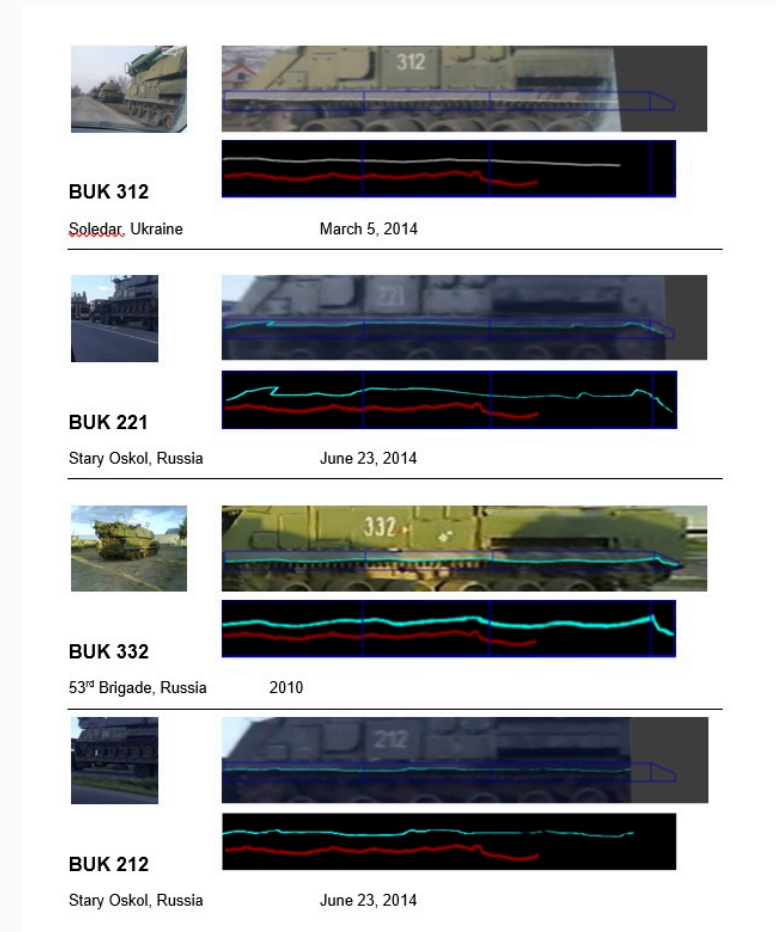
- Social media posts showed a Buk missile launcher in Eastern Ukraine
- Videos showed a Buk being transported on a truck
- Posts were geolocated to specific Ukrainian towns

Step 2: Identifying the Buk

- Close analysis of photos revealed unique markings
- Serial numbers and distinctive features matched
- The same Buk appeared in multiple locations

Step 3: Tracing its route

- Matched locations from videos to Google Maps
- Used Street View to verify landmarks
- Tracked the Buk from Russia to Ukraine and back



Matching distinctive features across multiple photos

How they did it: The Buk trail

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Reconstructed route of the Buk missile launcher

The crowdsourcing model

How Bellingcat operates

- Open call for evidence
- Community of contributors
- Public methodology
- Peer review through transparency

Why crowdsourcing worked

- One team couldn't analyze all available material
- Rapid response as evidence appeared
- Contributors brought specialized knowledge
- Multiple people checking each other's work
- Evidence saved before deletion

Bellingcat Launches the Ukraine Conflict Vehicle Tracking Project

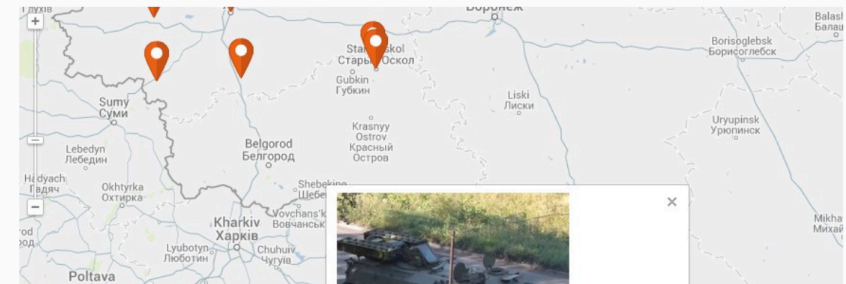
February 3, 2015 Russia Ukraine

Translations: [English \(UK\)](#)

Over the past few months, the Bellingcat team has been working on something slightly different from our previously published work. Now, we're pleased to launch a crowdsourced effort to track the movements of military vehicles both inside and in the vicinity of Ukraine, primarily to determine if equipment has been transferred across the border from Russia to Ukraine.

To kick-start the effort, we have used Daniel Romein's original research of last summer's military movements to build an initial data set for launch, supplemented with additional interesting sightings collected by the Bellingcat team through social media. The data is made available through [Silk](#), a beautiful data publishing and visualization platform.

You can access the Bellingcat Ukraine conflict vehicle tracking Silk here: <http://bellingcat-vehicles.silk.co/>



Impact and credibility

Media and official recognition

Bellingcat's findings were cited by international media worldwide, used by the official Joint Investigation Team, and confirmed by intelligence agencies

2018: Netherlands and Australia formally blamed Russia for the downing of MH17

2022: Dutch court found three men guilty of murder

Accountability in the information age

- Official investigations took years
- Bellingcat published findings within weeks
- Countered Russian disinformation
- Demonstrated power of open-source methods

Three men found guilty of murdering 298 people in shooting down of MH17

Court says Russia had overall control of the separatist forces in eastern Ukraine at the time when the plane was shot down

● [Russia-Ukraine war - latest news updates](#)



📷 A part of the wreckage is seen at the crash site of the Malaysia Airlines Flight MH17 in the Donetsk region of Ukraine. Photograph: Maxim Zmeyev/Reuters

the mag • MEDIA

Bellingcat, the NGO for which truth flows from open-source intelligence

By Lucas Minisini

Published on May 26, 2022, at 11:28 am (Paris), updated on May 26, 2022, at 11:28 am

The ethics and challenges of OSINT

Ethical considerations

Privacy concerns:

- Using personal social media posts
- Balancing transparency and protection

Verification burden:

- Risk of spreading misinformation
- What standards of proof are sufficient?

Trauma and well-being:

- Reviewing graphic content
- Secondary trauma for investigators
- Respecting victims and families

Technical challenges

The deepfake era:

- AI-generated content
- Increasingly sophisticated manipulation

Information overload:

- Too much data to analyze
- Automated disinformation campaigns

Platform dependence:

- Social media can delete evidence
- Platforms change access policies

Practice your OSINT skills!

What are OSINT challenges?

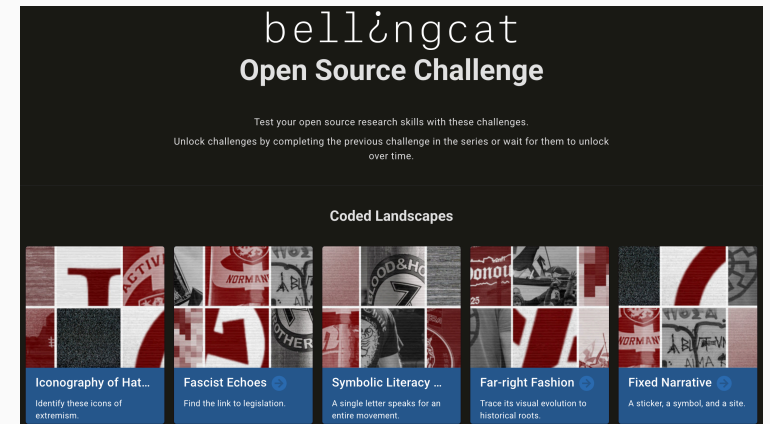
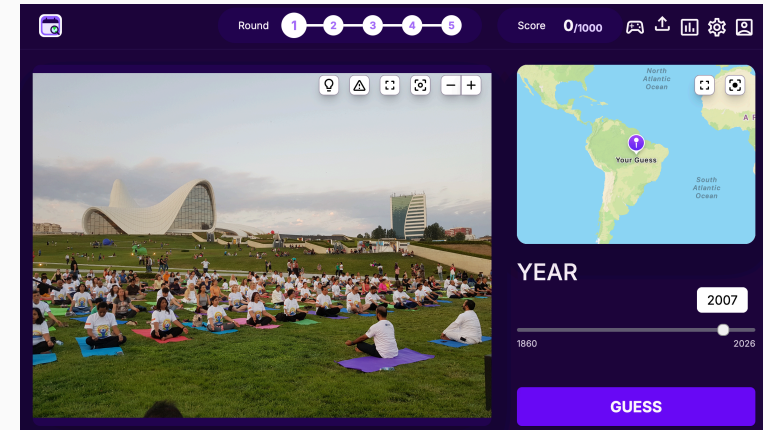
Online games and puzzles that train geolocation, verification, and open-source research skills. Ranging from "where was this photo taken?" to full investigations using satellite imagery, social media, and public records. Many are free and surprisingly addictive.

My picks

WhenTaken — Guess where *and* when a photo was taken. You're reading landscape, architecture, clothing, and signage all at once. Free, no account needed, immediately playable.

GeoGuessr — Drops you into Google Street View somewhere in the world. Builds visual intuition for landscape, architecture, and road markings. Free tier is limited but enough to get the idea.

Bellingcat Open Source Challenge — Monthly challenges designed by Bellingcat investigators. Goes well beyond geolocation: ship tracking, satellite imagery, flight data, image verification.



Discussion

Class discussion

Discussion questions

1. **OSINT and verification:** After reading about Bellingcat's MH17 investigation, what do you see as the biggest strengths and weaknesses of crowdsourced, open-source investigations?
2. **Trust and transparency:** Does showing your methods (like Bellingcat does) actually increase trust?
3. **Ethics and access:** What are the ethical boundaries of using people's social media posts for investigations?
4. **The future:** How might AI change OSINT journalism?

Coming up next week

How people understand numbers and visuals

Preparation:

1. Read the assigned papers on data visualization and numeracy
2. Think about examples of visualizations that confused or enlightened you
3. Consider how you personally process quantitative information